AKKA

Reliable AI for Every Industry
The Agentic Systems Platform

Akka vs. LlamaIndex

A comparison for teams building agentic AI

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LlamaIndex indexes and retrieves your data; Akka runs the agentic system that acts on it — and guarantees it. LlamaIndex is a strong RAG / data framework: connectors, parsing, indexing, and retrieval that give a model the right context. It is not an agentic runtime — production reliability, durable state, HA/DR, and runtime governance are yours to build, integrate, and operate. The Akka Agentic AI Platform delivers them and guarantees resilience and scalability, and LlamaIndex's retrieval layer can feed an Akka agent.

No SLA	99.9999%	4ms	Up to 90%
LLAMAINDEX PUBLISHED AVAILABILITY	AKKA PLATFORM SLA	AKKA DURABLE-STATE READS	CHEAPER TO OPERATE

DIMENSION	LLAMAINDEX	AKKA
What it is	A RAG / data framework for indexing and retrieval (plus LlamaCloud, a managed parsing/indexing service)	**The Akka Agentic AI Platform, which guarantees resilience and scalability**
Primary job	Connect, parse, index, and retrieve your data so an LLM has the right context	**Run, scale, persist, and govern agentic systems in production**
Agentic features	Workflows (event-driven orchestration), function-calling and ReAct agents, AgentWorkflow — a library you deploy and operate	**Native agents, durable memory, streaming, APIs, orchestration, and governance on one runtime**
Durable execution	Not built in — Workflows do not auto-checkpoint; durability needs an external integration (e.g., DBOS), replay-based and at-least-once	**Durable sharded in-memory state, event-sourced, replayable; 4ms reads / sub-10ms writes**
Availability SLA	No published numeric SLA; "uptime SLAs" referenced for the enterprise tier only	**99.9999% — entire platform, backed by indemnities**
HA/DR	Customer-owned (self-hosted) or per managed-service terms; no published active-active / RTO / RPO	**Active-active across regions; sub-1-minute RTO; zero-byte RPO**
Governance / EU AI Act	Observability and evaluation integrations; no inline enforcement, immutable ledger, pre-deployment classification, or sealed artifact	**Aspect-woven runtime enforcement + full pre-production governance**
Cost model	Consumption credits ($1 / 1,000 credits; ~3 credits/page; 10,000 free credits/mo) + you provision and operate everything else	**Shared compute; up to 90% lower infrastructure for the same workload, fixed annual fee**
Certifications	SOC 2 Type II reported for LlamaCloud	**19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)**
Vendor model	Venture-funded: $19M Series A (Mar 2025), ~$93M post-money; Databricks & KPMG strategic minority investments	**Profitable and self-funding; since 2007, 100,000+ deployments; Dell Technologies Capital is largest shareholder, customer, and AI partner**

LlamaIndex Is a Retrieval Framework; Akka Is the Agentic Platform

LlamaIndex is a data framework for retrieval-augmented generation: data connectors, document parsing (LlamaParse), indexing, and query/retrieval interfaces that give a model the right context. It does this well, and it has added agentic constructs — Workflows (event-driven orchestration), function-calling and ReAct agents, and AgentWorkflow. Those are libraries you import, deploy, and operate. The substrate a production agentic system runs on — a durable runtime, high availability, disaster recovery, and embedded governance — is not part of LlamaIndex; you build, integrate, and operate it yourself, and own every failure across the seams.

Capability	LlamaIndex	Akka
Data ingestion, parsing, indexing, retrieval (RAG)	Yes — its core strength	Not native (see complement note)
Agent constructs (tools, function calling, ReAct)	Library	Built in
Event-driven orchestration (Workflows)	Library	Built in
Durable execution / crash recovery	External integration (e.g., DBOS)	Built into the runtime
Durable memory	Bolt-on store	Built in, 4ms / sub-10ms
Real-time streaming	None native	Built in, backpressured, petabyte-scale
HTTP / gRPC API layer	None	Built in
Runtime governance / policy enforcement	None	Inline, runtime-embedded
Pre-production governance	None	Classification, sign-offs, sealed posture

Availability, Durability, and Disaster Recovery

LlamaIndex does not publish a numeric availability SLA; "uptime SLAs" are referenced only generically for the enterprise tier, with no stated percentage, RTO, or RPO. More important for agentic AI is durable execution. LlamaIndex Workflows do not automatically snapshot state — by design, to avoid overhead — so a workflow cannot recover from a crash on its own. Durable execution is available only through an external integration such as DBOS, which journals step completions to a database and replays on restart; recovery is at-least-once, so steps can run more than once and the developer must make them idempotent.

Metric	LlamaIndex	Akka
Availability SLA	None published (numeric)	99.9999%
Allowed downtime / year	Not specified	~31 seconds
Durable state	External integration; replay-based, at-least-once	Event-sourced, built in; exactly-once recovery
State latency	Per external store	4ms reads / sub-10ms writes
RTO / RPO	Not published	Sub-1-minute RTO / zero-byte RPO
HA/DR	Customer-owned / per managed terms	Active-active across regions
SLA scope	—	The entire platform, backed by indemnities

Akka provides durability and fault tolerance as part of the runtime: state is event-sourced in durable sharded in-memory storage, replayable from its own journal, with active-active HA/DR, sub-1-minute RTO, and zero-byte RPO under a 99.9999% SLA. Eighteen years and 100,000+ production deployments stand behind it.

Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not list price. The drivers are actor concurrency (~10 trillion tokens/core/year vs ~2 trillion for comparable solutions; ~80% less compute than Python-based frameworks), shared compute, and micro-checkpointing. Manulife reported up to 300% more concurrency and 30–50% faster processing after porting Python-based systems to Akka.

LlamaCloud bills by consumption — $1 per 1,000 credits, roughly 3 credits per page for cost-effective parsing, with 10,000 free credits per month — and that meter covers parsing, extraction, and indexing: the retrieval layer. The runtime to run agents in production (compute, memory, streaming, APIs, observability, governance, HA/DR) is provisioned and paid for separately, on top. Akka runs all of it on one shared-compute runtime for a fixed annual fee finance can forecast.

Governance and the EU AI Act

LlamaIndex provides observability and evaluation: instrumentation, tracing integrations, and evaluation tooling to measure retrieval and agent quality. It does not provide AI-governance enforcement — no real-time policy enforcement, no decision explainability, no human pause/override of a running process, no immutable interaction ledger, no pre-deployment classification, and no sealed audit artifact.

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

High-risk AI carries a 10-year logging-retention obligation (Art. 72), enforceable since Feb 2025 (prohibited) / Aug 2025 (high-risk).

How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL human control; atomic PII scrub-with-explain; pre-deployment classification against **190 regulations and 1,040 controls** (671 carrying a financial penalty); multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Governance that LlamaIndex would leave a customer to source and bolt on, Akka enforces inline.

Two Lifecycles, One Certified System

Building on LlamaIndex means engineers writing retrieval and workflow code; there is no built-in path for a risk officer or compliance lead to contribute, and no governance lifecycle. Akka runs two independent lifecycles on one platform via **Akka Specify**:

BUILD LIFECYCLE
FUNCTIONAL CONTRACT
"Rank incoming ER patients by acuity and route the top three to a clinician."
Product · developers · ML engineers · domain experts
v1.4 · versioned · tested

GOVERN LIFECYCLE
SAFEGUARD CONTRACT
"Block prohibited practices under EU AI Act Article 5; notify regulators within 24h of any incident."
Risk · security · compliance
v2.1 · versioned & tested independent of the build

→

Akka Specify
AI-assisted authoring
generates · tests · runs

→

ONE CERTIFIED AI SERVICE
Built, governed, and running
- Agents, tools, orchestration, memory, APIs, streaming, UI
- Guardrails, sanitizers, HITL/HOTL, evaluations, halts
- Interaction, evidence, and causal logging

Akka Verify ↻ validates the running system against both specs and fine-tunes the AI from production data.

The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow LlamaIndex has no equivalent for.

Real-Time Streaming at Petabyte Scale

LlamaIndex has no streaming engine; real-time pipelines are provisioned separately. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: Risk, Compliance, and Accountability

Buyer concern	LlamaIndex	Akka

Certifications & audits	SOC 2 Type II reported for LlamaCloud	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Scope of accountability	The retrieval layer; you integrate and operate the runtime, agents, and governance	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard cloud terms	Availability and data-integrity guarantees backed by contractual indemnities
Track record & funding model	Venture-funded: $19M Series A (Mar 2025), ~$93M post-money; Databricks and KPMG strategic minority investments	Profitable and self-funding; since 2007 and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner
Budget predictability	Consumption credits that scale with load	Fixed annual fee finance can forecast

The decision is scope and accountability: LlamaIndex gives you a best-in-class retrieval layer to feed an agent; Akka gives you the platform that runs and guarantees the agent.

> **Akka complements your retrieval layer.** Akka is not a vector database or a semantic knowledge layer, and does not aim to be. LlamaIndex's strength — parsing, indexing, and retrieving your data — is real and complementary. A LlamaIndex retrieval pipeline can supply context to an Akka agent, which then runs that agent as a durable, highly available, governed production system. Retrieval feeds the agent; Akka runs it.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We already use LlamaIndex for RAG. Why add Akka?

LlamaIndex is strong for connecting, indexing, and retrieving your data. To put an agent into production you also need a durable runtime, high availability and disaster recovery, durable memory, streaming, and runtime governance — which LlamaIndex does not provide. Akka runs the agent as a guaranteed production system, and a LlamaIndex retrieval pipeline can feed it.

LlamaIndex has Workflows and agents now. Isn't that an agentic runtime?

Workflows and agents are libraries you deploy and operate yourself. They do not auto-checkpoint state, so crash recovery requires an external integration such as DBOS journaling to a database, with at-least-once semantics you design around. Akka provides durable, event-sourced execution, active-active HA/DR, and a 99.9999% SLA as part of the runtime.

Can we add governance on top of LlamaIndex?

You can add observability and evaluation tools, but the EU AI Act expects enforcement inline to the runtime: immutable records witnessed as they happen, human override on running processes, and pre-deployment classification. Tools that observe and evaluate cannot gate a deployment or seal an audit artifact. Akka embeds all of this and covers pre-deployment governance.

Is LlamaIndex cheaper because the framework is open source?

The OSS framework is free, but production means LlamaCloud's consumption credits for parsing and indexing plus the separate runtime you provision and operate for everything else. Akka's shared-compute model is up to 90% cheaper to operate than the equivalent assembled stack, on a fixed annual fee.

Sources

LlamaIndex product / RAG: llamaindex.ai · developers.llamaindex.ai/python/framework/ — open-source data framework for RAG (ingest, index, retrieve); "AI Agents for Document OCR + Workflows" (Jun 2026)

LlamaIndex agentic constructs: developers.llamaindex.ai/python/llamaagents/workflows/ — Workflows (event-driven), AgentWorkflow, function-calling / ReAct agents

Workflows durability: developers.llamaindex.ai/python/llamaagents/workflows/dbos/ — Workflows do not auto-snapshot; durable execution via DBOS journaling, replay-based, at-least-once (idempotency required) (2026)

LlamaCloud pricing: llamaindex.ai/pricing · developers.llamaindex.ai/python/cloud/general/pricing/ — $1 per 1,000 credits; ~3 credits/page (cost-effective); 10,000 free credits/month

LlamaIndex SLA / security: no published numeric availability SLA; "uptime SLAs" referenced for enterprise tier only; SOC 2 Type II reported for LlamaCloud

LlamaIndex funding: prnewswire.com / crunchbase.com — $19M Series A led by Norwest with Greylock, Mar 4 2025, ~$93M post-money, $27.5M total disclosed; Databricks Ventures + KPMG strategic minority investments (May 2025)

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300%

more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka platform: 99.9999% availability, active-active HA/DR, sub-1-min RTO, zero-byte RPO (contractual indemnities); 4ms reads / sub-10ms writes; 190 regulations / 1,040 controls / 671 with a financial penalty; 100,000+ deployments / since 2007 / 52 banks; profitable; Dell Technologies Capital